IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 4 July 2024, accepted 24 July 2024, date of publication 2 August 2024, date of current version 14 October 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3437247

APPLIED RESEARCH

Short-Term Power Load Forecasting Based on DE-IHHO Optimized BiLSTM

XUELEI LIU [1], ZIQI MA [1], HANRUI GUO [1], YEDONG XU [1], AND YINGLI CAO [1,2]
[1]College of Information and Electrical Engineering, Shenyang Agricultural University, Shenyang 110866, China
[2]Liaoning Key Laboratory of Intelligent Agricultural Technology, Shenyang 110866, China

Corresponding author: Yingli Cao (caoyingli@syau.edu.cn)

This work was supported by the Key Research Project of Liaoning Provincial Department of Education under Grant JYTZD2023123.

ABSTRACT Accurate short-term power load forecasting is the key to determining the grid company's dispatch plan and system operation mode. Aiming at the problem of low prediction accuracy due to the difficulty in selecting hyperparameters of BiLSTM, a hybrid parallel Harris hawk optimization algorithm (DE-IHHO) is proposed to choose the optimal hyperparameters of BiLSTM to improve the prediction accuracy of the model. In this paper, several load forecasting models are tested and BiLSTM with better performance is chosen as the baseline model. Aiming to solve the problem of the complex selection of hyperparameters for BiLSTM, the Harris Hawk Optimization (HHO) algorithm is used to obtain better hyperparameter combinations and improve prediction accuracy. To further explore the optimal hyperparameter combinations of BiLSTM, running in parallel with differential evolutionary algorithm (DE) to enhance the search diversity, adopting chaotic dyadic learning strategy and mutation operation strategy to improve the global search capability of HHO, and finally smoothing the optimal solution to reduce the influence of abnormal solutions. The results show that the convergence speed and optimization ability of DE-IHHO are significantly improved, and the BiLSTM optimized in this way improves considerably in all three metrics of MAE, MAPE, and RMSE, proving this prediction model's effectiveness.

INDEX TERMS Short-term load forecasting, bidirectional long and short-term memory neural network (BiLSTM), hybrid parallel Harris Hawk optimization algorithm (DE-IHHO), chaotic dyadic learning strategy, variational manipulation strategy.

I. INTRODUCTION

The development of the new power system manifests itself in the access to a large number of renewable energy sources, such as wind power and photovoltaic on the supply side, and in the access to a large number of flexible loads, such as electric vehicles and central air-conditioning on the user side. Accurate short-term power load forecasting is the basis for realizing demand response rational scheduling of flexible loads to consume renewable energy and avoid energy waste fully [1]. The research on load forecasting has been developed from single-feature input to multi-feature input, and the literature [2], [3] compares the single-feature input model with the multi-feature model considering meteorological factors, which shows that the multi-feature input has a higher prediction accuracy and has become the mainstream of the research nowadays. Short-term power load forecasting methods that consider multi-feature inputs are mainly categorized into traditional and machine learning methods.

Traditional load forecasting methods include exponential smoothing [4], Kalman filtering [5], etc. Traditional forecasting methods are used for load forecasting by analyzing the correlation between load data. These statistical methods have the advantage of simplicity and speed. Still, they do not apply to complex nonlinear systems and have a small range of applicability, so the traditional methods are not applicable in this study.

Artificial neural networks are the research hotspot of machine learning-based power load forecasting methods. Artificial neural networks have a solid ability to fit data, but it is not easy to find the optimal global solution [6], [7]. This resulted in Recurrent Neural Networks (RNN), which have

The associate editor coordinating the review of this manuscript and approving it for publication was Sotirios Goudos.

a temporal memory capability to retain relevant information in the network [8], solving the problem of traditional neural networks dealing with time series data. However, RNNs are prone to gradient explosion and gradient vanishing during training. In this regard, literature [9], [10] proposed a Long Short-Term Memory Network (LSTM). LSTM solves the problem of gradient vanishing in the RNN structure and enhances the long-term memory ability, better reflecting its excellent performance in long-time sequences. However, LSTM is still deficient in mining the deep relationship of load features.

Bidirectional Long Short-Term Network (BiLSTM) models have advantages in obtaining contextual information on time series data [11], [12]. Literature [13] used BiLSTM for short-term power load forecasting and achieved excellent results. However, BiLSTM has some drawbacks, and incorrect parameter selection can lead to degraded model performance or unstable training. Literature [14], [15] overcame the problem of difficult parameter selection by optimizing the BiLSTM model with a particle swarm, obtained the optimal hyperparameters, and achieved better prediction results. The genetic algorithms used in the literature [16], [17], [18] optimize BiLSTM networks for prediction, and their accuracy is significantly improved. However, when the search space is too complex and limited by the search capability of intelligent algorithms, the solution process quickly falls into the local optimum, and it becomes essential to improve the global search capability of intelligent algorithms. Literature [19], [20], [21] used hybrid intelligent algorithms for power load forecasting to increase the diversity of search and improve the prediction accuracy compared to single intelligent algorithms.

To address the above issues, this paper tests a variety of load forecasting models, and the results show that BiLSTM with bi-directional structure predicts better. Therefore, BiLSTM is chosen as the baseline model in this study. HHO is used to solve the problem of difficult hyperparameter selection of BiLSTM, and the results show that the accuracy of the HHO-BiLSTM prediction model is improved. To enhance the global search capability of HHO to mine better hyperparameter combinations further, this paper adopts parallel computing with DE to enhance the search diversity, adopts chaotic dyadic learning strategy and variational operation strategy to improve the global search capability of HHO, and finally smoothes the optimal solution to reduce the influence of abnormal solutions. The results show that the convergence speed optimization ability of DE-IHHO is significantly improved relative to HHO, which mines out better hyperparameter combinations of BiLSTM and substantially improves the prediction accuracy.

II. BASED ON THE BiLSTM LOAD FORECASTING MODEL CONSTRUCTION

LSTM networks are improved from recurrent neural networks by introducing a "gating" mechanism, which

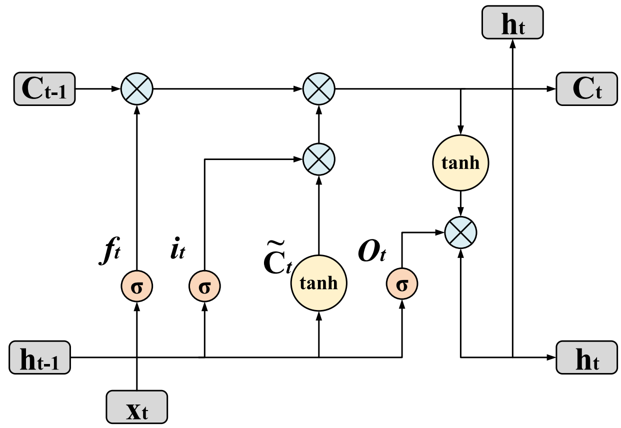


FIGURE 1. LSTM unit structure diagram.

selectively remembers or forgets information to deal with long-term dependencies more effectively [22]. The LSTM structure is shown in Figure 1. In LSTM networks, the forgetting gate f_t, the input gate i_t, and the output gate O_t are added inside each neuron, x_t and h_{t-1} are the inputs, and a stream of information representing the long-term memory is added [23]. Where W and b are the weight matrix and bias matrix, respectively, $\sigma(\cdot)$ is the sigmoid activation function, C_t is the cell state at the current moment, and h_t It is the hidden layer output vector.

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) d \tag{1}$$
$$C_t = f_t \bullet {}_{t-1} + g_t \bullet {}_t \tag{2}$$
$$h_t = \tanh\left(f_t \bullet {}_{t-1} + g_t \bullet {}_t\right) \bullet {}_t \tag{3}$$
$$\tilde{C}_t = \tanh\left(W_g \bullet h_{t-1}, x_t] + b_g\right) \tag{4}$$
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{5}$$
$$O_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{6}$$

The BiLSTM network is a model that combines bidirectional propagation and an extended short-term memory network (LSTM). BiLSTM supports the bi-directional communication of information to mine further the intrinsic relationships between current, past, and future data. Literature [24] used the BiLSTM network to improve further the model's prediction accuracy and the utilization of feature data. The BiLSTM network model is shown in Figure. 2. The expression of the BiLSTM network is.

$$h_f = \mathrm{LSTM}\left(x_t, h_{f-1}\right) \tag{7}$$
$$h_b = \mathrm{LSTM}(x_t, h_{b-1}) \tag{8}$$
$$h_t = w_t h_f + v_t h_b + b_t \tag{9}$$

In Eqs. (7)-(9) h_f is the forward layer output, h_b is the inverse layer output, w_t is the forward hidden layer output weight, v_t is the inverse hidden layer output weight, and b_t is the hidden layer bias. In this study, x_t is a region's temperature, humidity, rainfall, and load history data as input, and h_t is the load prediction value as output.

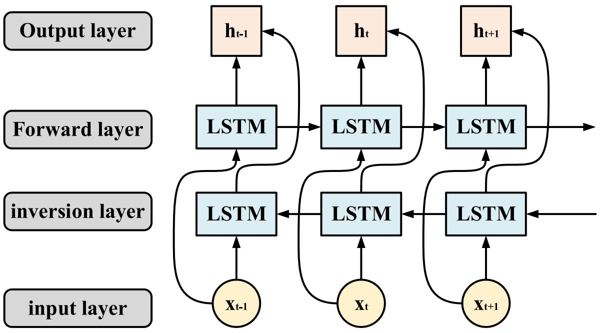


FIGURE 2. BiLSTM cell structure diagram.

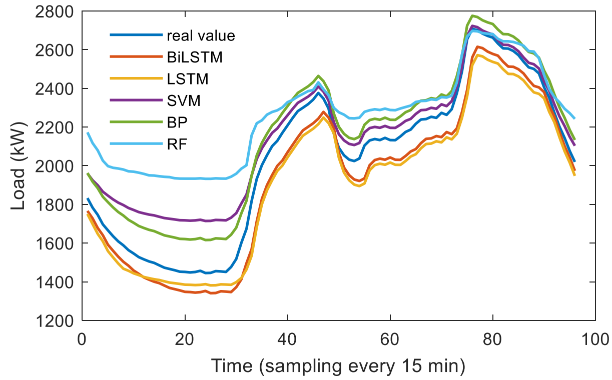


FIGURE 3. Comparison of different model prediction results.

A. LOAD FORECASTING MODEL EVALUATION INDICATORS

In this paper, three evaluation indexes, Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error (RMSE), are used to validate the effectiveness of the model.

$$MAE(x_j, \hat{x}) = \frac{1}{m}\sum_{1}^{m} | x_j - \hat{x} | \tag{10}$$

$$MAPE(x_i, \hat{x}) = \frac{1}{m}\sum_{1}^{m} \frac{|x_j - \hat{x}|}{|x_j|} \tag{11}$$

$$RMSE(x_j, \hat{x}) = \sqrt{\frac{1}{m}\sum_{1}^{m}(x_j - \hat{x})^2} \tag{12}$$

where x_j Is the actual value of load data, $\hat{x}$ It is the predicted value of load data, m is the number of data, and the smaller the value of MAE, MAPE, and RMSE, the better, representing the higher accuracy of the prediction model.

B. ANALYSIS OF BiLSTM LOAD FORECASTING RESULTS

The experimental data were selected from the electricity consumption data collected every 15 minutes in January 2023 in a northeastern region, as well as meteorological data. The data at a specific moment are shown in Table 1, and the average daily electricity consumption in the area is 48906.25 kW·h. The prediction model takes maximum temperature, minimum temperature, average temperature, relative humidity, rainfall, and historical load data as inputs and load prediction values as outputs. The number of training times is set to 100, the learning rate is 0.0384, and the number of hidden layer neurons is 5.

Because BiLSTM comprehensively considers the characteristics of the positive and negative information law of the time series data, as shown in Fig. 3 BiLSTM is closer to the real value compared with the long-short-term network (LSTM), BP neural network, Support Vector Machine (SVM), and the Random Forest Model (RF), and as shown in Table 2 BiLSTM outperforms the other forecasting models in the three evaluation indexes of MAE, RMSE, and MAPE model, but the error of BiLSTM still reaches 2476.4KW·h accounting for 5.1% of the total daily electricity consumption, which does not satisfy the standard that the error of load forecasting in China generally does not exceed 3%. Since the hyperparameters of BiLSTM are empirically given as not optimal, resulting in significant prediction errors, this paper continues to explore HHO to optimize the hyperparameters of BiLSTM to improve load prediction accuracy.

III. HHO-BASED BiLSTM HYPERPARAMETER SELECTION

BiLSTM has a significant influence on the prediction results of each parameter of the model when performing load prediction and the results of hyperparameters such as the number of iterations, the number of hidden neurons, the learning rate, etc., which are searched according to the previous experience, will lead to overfitting or underfitting of the model. To solve the above problems, this study applies HHO to optimize the hyperparameters to improve the BiLSTM model's prediction performance.

A. HHO SOLVING STRATEGY

In 2019, Heidari et al. proposed a new population intelligence optimization algorithm-Harris Hawks Optimization (HHO). Harris Hawks first performs a global search for the prey, determines whether to perform a local search by the energy factor, selects the hunting method according to the prey, and finally captures the prey [25], [26], and the solution strategy is as follows:

(1) Global search strategy

The Harris Hawk is randomly distributed in space and searches for prey in space through two strategies. The location update strategy is shown below:

$X(t+1)$ represents the position of the hawk in $(t+1)$ iterations, X_{rabbit} represents the position of the prey q, r_1, r_2, r_3, r_4 represents the random numbers within (0,1) that are updated in each iteration, UB, and LB are the upper and lower bounds, respectively, X_{m} represents the average position of the hawk, and N is the size of the population.

$$X(t+1) = \begin{cases} X_{\text{rand}}(t) - r_1 |X_{\text{rand}}(t) - 2r_2 X(t)| \\ \quad q \geq 0.5 \\ (X_{\text{rabbit}}(t) - X_{\text{m}}(t)) - r_3 (LB + r_4 (UB - LB)) \\ \quad q < 0.5 \end{cases} \tag{13}$$

TABLE 1. Part test data.

dates	highest temperature℃	Minimum temperature℃	average temperature℃	relative humidity (average)	quantity of rainfall（mm）	burden (KW)
2023/1/5	9.2	5.1	6.9	78	2.9	3295.41
2023/1/6	11	6.3	8.2	92	3.3	3167.59
2023/1/7	12.4	9.4	10.5	80	0	3140.76
…	…	…	…	…	…	…

TABLE 2. Comparison of the results of the evaluation indicators of different models.

Model name	MAE(KW)	MAPE(%)	RMSE(KW)
BiLSTM	103.18	5.28	112.66
LSTM	120.01	5.93	130.49
SVM	108.99	6.54	141.68
BP	116.55	6.32	121.24
RF	218.09	12.88	272.48

$$X_m(t) = \frac{1}{N}\sum_{i=1}^{N} X_i(t) \tag{14}$$

(2) Basis for switching between global and local search strategies

The global and local search strategy of HHO is controlled through the energy factor E with the following expression.

$$E = 2E_0\left(1 - \frac{t}{T}\right) \tag{15}$$

E_0 denotes the initial energy of variation within (-1,1), T denotes the maximum number of iterations, and t represents the current iteration. when $|E| \geqslant 1$, a global search strategy is adopted; when $|E| < 1$, a local search strategy is adopted.

(3) Local search strategy

The Harris Hawk selects a localized search strategy using a prey escape factor, r, and an energy factor, E, as follows:

a. Soft roundup

When |E|⩾0.5 and r⩾0.5, the prey has enough energy to escape, and the Harris Hawk ultimately captures the prey by consuming its power, with the positional update expression as follows:

$$X(t+1) = \Delta X(t) - E\left|JX_{\text{rabbit}}(t) - X(t)\right| \tag{16}$$

$$\Delta X(t) = X_{\text{rabbit}}(t) - X(t) \tag{17}$$

$$J = 2\left(1 - r_5\right) \tag{18}$$

$\Delta X(t)$ Is the difference between the prey position and the current individual position vector, J is the prey's random jump strength, and r_5 is a random number in (0,1).

b. Hard Surround Hunting

When $|E| < 0.5$, $r \geqslant 0.5$, the prey does not have enough energy to escape, and the Harris Hawk encircles the prey directly with the position update expression as follows:

$$X(t+1) = X_{\text{rabbit}}(t) - E\left|\Delta X(t)\right| \tag{19}$$

c. Tired speed swooping soft roundups

When $r < 0.5$, $|E| \geqslant 0.5$, the prey still has enough energy to escape successfully, the Harris Hawk first performs a swooping raid on the prey, and if the raid fails, it employs a stochastic wandering strategy with the following expression:

$$Y = X_{\text{rabbit}}(t) - E\left|JX_{\text{rabbit}}(t) - X(t)\right| \tag{20}$$

$$\mathrm{Z} = \mathrm{Y} + \mathrm{S} \times \mathrm{Levy}(D) \tag{21}$$

$$X(t+1) = \begin{cases} Y & \text{if } F(Y) < F(X(t)) \\ \mathrm{Z} & \text{if } F(\mathrm{Z}) < F(X(t)) \end{cases} \tag{22}$$

D is the problem dimension, S is a $1 \times D$ dimensional random vector, and the Levy expression is as follows:

$$\mathrm{Levy}(x) = 0.01 \times \frac{u \times \sigma}{|v|^{\frac{1}{\beta}}},$$
$$\sigma = \left(\frac{\Gamma(1+\beta) \times \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\left(\frac{\beta-1}{2}\right)}}\right)^{\frac{1}{\beta}} \tag{23}$$

u, v are random values within (0,1), and β is a constant value.

d. Tired speed dive hard roundups

When $r < 0.5$ and $|E| < 0.5$, the prey has insufficient energy, Harris's hawk adopts hard encirclement for predation, and if the raid fails, it adopts a random wandering strategy with the following expression:

$$X(t+1) = \begin{cases} Y & \text{if } F(Y) < F(X(t)) \\ \mathrm{Z} & \text{if } F(\mathrm{Z}) < F(X(t)) \end{cases} \tag{24}$$

$$Y = X_{\text{rabbit}}(t) - E\left|JX_{\text{rabbit}}(t) - X_m(t)\right| \tag{25}$$

$$\mathrm{Z} = \mathrm{Y} + \mathrm{S} \times \mathrm{Levy}(D) \tag{26}$$

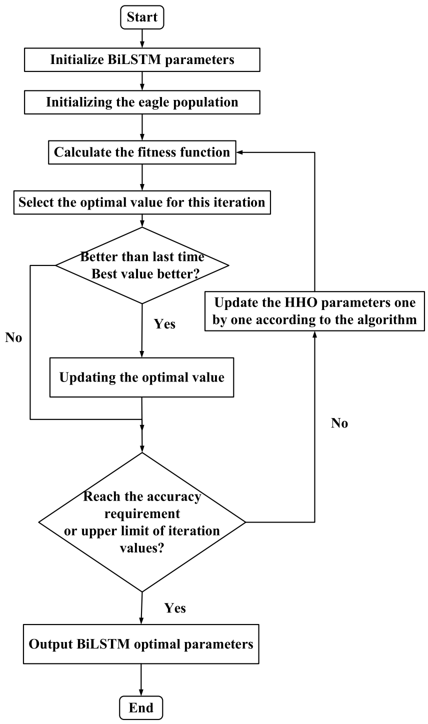


FIGURE 4. HHO algorithm parameter optimization flow chart.

B. DESIGN OF HHO-BASED HYPERPARAMETER OPTIMIZATION PROCESS

The number of initial HHO hawks is set to 20; the number of iterations is 10; the algorithm's performance is tested using RMSE as a fitness function.

(1) Set the initial hyperparameters of BiLSTM: the number of neurons in the hidden layer, training times, and the learning rate.

(2) Initialise the eagle population

(3) Calculate the root-mean-square error of the corresponding BiLSTM model for the location of each eagle as the fitness function.

(4) Sort the eagles according to the fitness function and select the optimal value for this iteration.

(5) Compare the highest value of this adaptation with the historical optimal value and update the global optimal parameters if the adaptation is improved.

(6) Judge whether to meet the number of iterations or accuracy requirements. If the requirements are met, then output BiLSTM optimal parameters.

The specific flow of model calculation is shown in Figure 4

C. ANALYSIS OF HHO-BiLSTM LOAD FORECASTING RESULTS

The data were selected in Table 1, and the results of the hyperparameter optimization selection are shown in Table 3. The number of training times is kept unchanged from the original, the learning rate is changed to 0.02556, and the number of neurons in the hidden layer is changed to 10. The error is reduced to 1473.5 KW·h 3% of the total daily electricity

TABLE 3. BiLSTM parameter optimization results.

Optimization parameters	Epochs	Hidden layer neurons (pcs)	learning rate
optimal outcome	100	10	0.02556

TABLE 4. Comparison of BiLSTM and HHO-BiLSTM model evaluation index results.

Model name	MAE(KW)	MAPE(%)	RMSE(KW)
BiLSTM	103.18	5.28	112.66
HHO-BiLSTM	61.40	3.07	74.67

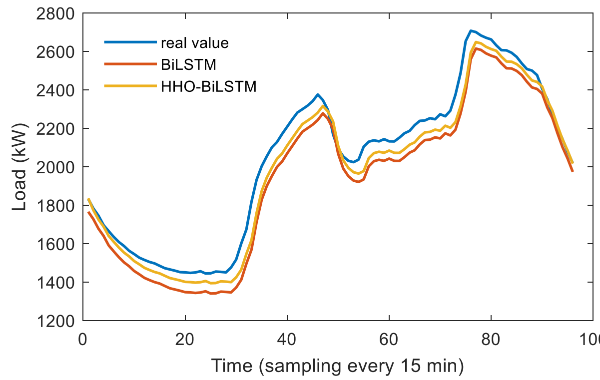


FIGURE 5. Comparison of prediction results of HHO-BiLSTM model.

consumption 2% reduction in error compared to the unoptimized BiLSTM. As shown in Table 4, after applying HHO optimization, the BiLSTM model MAE, MAPE, and RMSE are improved by 41.78KW, 2.21%, and 37.99KW, respectively, and the prediction effect is significantly enhanced. This indicates that the prediction model's performance improved after selecting hyperparameters through HHO. Still, better hyperparameters may not be mined due to the limitation of insufficient global search capability of HHO, and there is a possibility of mining better hyperparameter combinations through its improvement to improve the prediction accuracy further.

IV. BiLSTM HYPERPARAMETER SELECTION BASED ON DE-IHHO

Aiming at the insufficient global search capability of the HHO algorithm, which leads to the selection of hyperparameters that may not be optimal and thus leads to lower prediction accuracy, this paper constructs a hybrid parallel Harris Hawk optimization algorithm (DE-IHHO) to solve the above problems. Hybrid parallel computation with HHO and DE increases search diversity and provides a solution strategy for DE. At the same time, HHO is improved by adopting an adversarial learning strategy and logistic chaotic mapping sequences for its initial population to generate uniformly distributed initial solutions so that the population is uniformly distributed and diverse; the global search strategy of HHO is updated by integrating the variation operator (DE/current-to-best/1) to reduce the risk of falling into the local optimum. Finally, the optimal solution is smoothing to reduce the influence of abnormal solutions.

A. DE SOLVING STRATEGY

Differential Evolution Algorithm (DE) is a population-based stochastic search algorithm that simulates the natural evolutionary process through mutation, crossover, and selection to move its population towards the global optimum [27], [28], [29], [30]. The specific process is as follows:

(1) Variation strategy

Three different individuals X_{r1}, X_{r2}, X_{r3} are selected from the population to generate a new solution V_i From the current solution in the search space with the following expression:

DE/best/1:

$$V_i = X_{\text{best}} + F\,(X_{r1} - X_{r2}) \tag{27}$$

DE/current-to-best/1:

$$V_i = X_i + F\,(X_{\text{best}} - X_{r1}) + F\,(X_{r2} - X_{r3}) \tag{28}$$

X_{best} is the optimal solution, and F is the variation factor within [0, 1]. In this paper, DE/best/1 is used as the variation operator, accelerating the convergence speed and maintaining the population diversity.

(2) Cross-cutting strategies

A test vector U_i is generated from the corresponding mutation solution vector V_i and the target vector X_i, with the following expression:

$$U_{ij} = \begin{cases} V_{ij} & \text{if rand }(0,1) \le CR \text{ or } j = j_{\text{rand}} \\ X_{ij} & \end{cases} \tag{29}$$

where CR is the crossover rate, j_{rand} is a random value on [1, D], and D is the problem dimension.

(3) Selection strategy

Choosing between X_i or U_i Based on their fitness values, the fitter for the next iteration is selected based on.

$$X_i\,(t+1) = \begin{cases} U_i\,(t) & \text{if } f(U_i(t)) \le f(X_i(t)) \\ X_i(t) & \text{otherwise} \end{cases} \tag{30}$$

B. IMPROVED HHO WITH SMOOTHING

1) CHAOTIC DYADIC LEARNING INITIALISING POPULATIONS

a: LOGISTIC CHAOTIC MAPPING SEQUENCE

The logistic chaotic mapping sequence has the characteristics of traversal and non-repeatability. The traversal of chaos means that it can cover different regions in the solution space faster. At the same time, its non-repeatability ensures that the search process does not visit the same solution repeatedly and is less likely to fall into local optima [31]. Therefore, the use of chaotic mapping sequences to adjust the stochastic parameters of HHO can help to improve the performance of the HHO algorithm with the following expression:

$$lr\,(t+1) = \mu \cdot lr\,(t) \cdot (1 - lr\,(t)) \tag{31}$$

where $\mu = 4$, $lr\,(1) \in (0,1)$, $lr\,(1) \neq 0.25, 0.5, 0.75, 1$. Replace the stochastic parameters of the HHO with $lr\,(t)$.

b: CONTRASTIVE LEARNING STRATEGIES

Opposition-based learning (OBL) strategies aim to simultaneously generate the "opposite" positions of a solution during the initialization phase of a population to cover most of the regions in the feasible domain, thus increasing the diversity of initial solutions [32]. Even without prior knowledge, using OBL can still result in more suitable starting candidates and increase the probability of detecting better regions. For solution X_{ij}, its opposing solution X^*_{ij} It can be identified as the following.

$$X^*_{ij} = lb_j + ub_j - X_{ij} \tag{32}$$

where j=1,2,…, D (dimensions of the specific problem), i=1,2,…, N (total number of eagles) values, and ub and lb represent the upper and lower bounds of the search space, respectively.

2) IMPROVING THE HHO GLOBAL SEARCH STRATEGY

DE/current-to-best/1 is a DE variant operation strategy that combines the current solution and the historical optimal solution to generate a new candidate solution, which is introduced in the global search phase of HHO, thereby increasing the exploration capability of the algorithm and decreasing the risk of falling into a local optimum [31], with the following improved formulation:

$$X_i\,(t+1) = \begin{cases} X_i(t) + F(\mathrm{X}_{\text{rabbit}}(t) - X_{r1}(t)) + F(X_{r2}\,(t) \\ \quad -X_{r3}\,(t))q \ge 0.5 \\ (\mathrm{X}_{\text{rabbit}}(t) - \mathrm{X}_{\text{m}}(t)) - r_4(LB + r_5(UB - LB)) \\ \quad q < 0.5 \end{cases} \tag{33}$$

where r is a (0,1) random number.

$X_i\,(t+1)$ and $X_i\,(t)$ denote the position of the Harris hawk at the t+1st and the iteration, respectively, $\mathrm{X}_{\text{rabbit}}(t)$ denotes the position of the prey at the tenth iteration, and $X_{\text{m}}(t)$ denotes the average position of the Harris hawk at the tenth iteration. r and q are random numbers in (0,1), and UB and LB are the upper and lower bounds of the dimensional space, respectively.

3) DE-IHHO BASED SMOOTHING

To reduce the control function fluctuations and mutation points in the optimal solution, the optimal solution of DE-IHHO is smoothed. The average value based on three consecutive points is used to reduce the jumps, and the smoothing formula at point k is.

$$\bar{u}_k = \begin{cases} \lambda u_{k-1} + (1-\lambda)u_{k+1}, \\ \text{if} \operatorname{sign}(u_k - u_{k-1})\operatorname{sign}(u_{k+1} - u_k) < 0 \\ u_k, \quad \text{else} \end{cases} \tag{34}$$

$\bar{u}_k$ represents the value of u_k After smoothing.

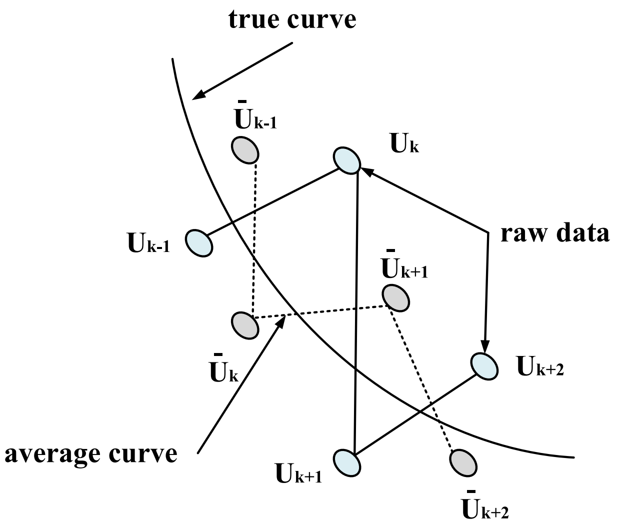


FIGURE 6. Application of smoothing technique in a control variable.

TABLE 5. BiLSTM parameter optimization results.

Optimization parameters	Epochs	Hidden layer neurons (pcs)	learning rate
optimal outcome	200	12	0.0008

TABLE 6. Comparison of evaluation index results of HHO-BiLSTM and DE-IHHO-BILSTM model.

Model name	MAE(KW)	MAPE(%)	RMSE(KW)
HHO-BiLSTM	61.40	3.07	74.67
DE-IHHO-BiLSTM	37.01	1.84	47.85

C. DESIGN OF HYPERPARAMETER OPTIMIZATION PROCESS BASED ON DE-IHHO

(1) Set BiLSTM initial network parameters: number of hidden layer neurons, training times, and learning rate.

(2) Initialise the hawk population.

(3) Use logistic chaotic sequence instead of random parameters of HHO.

(4) Calculate the opposing populations of all initial eagles by DBL strategy.

(5) Select the most suitable eagle as the initial population based on the fitness of the initial eagles to the opposing population.

(6) Update half of the new population by IHHO with the position of the eagle and the other part due to DE with the position of the other half of the eagle.

(7) Smooth the output.

(8) Merge the two parts of the updated sub-populations and update the optimal solution.

(9) Determine whether the number of iterations or accuracy requirements are met and output the BiLSTM optimal parameters if the requirements are met.

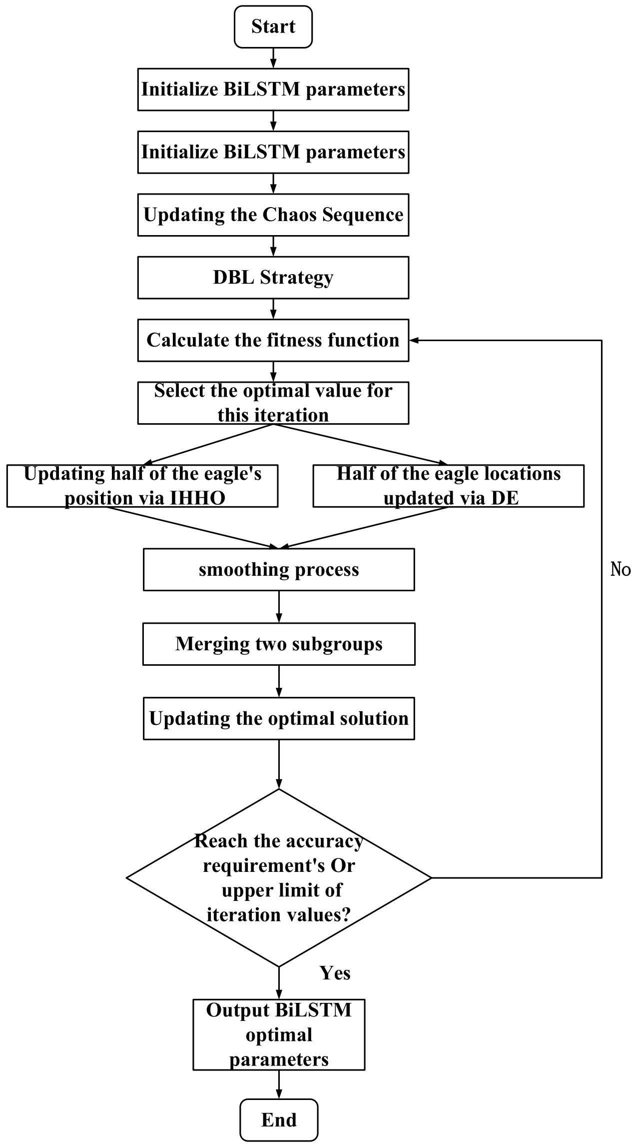


FIGURE 7. DE-IHHO algorithm parameter optimization flow chart.

D. ANALYSIS OF LOAD FORECASTING RESULTS BASED ON DE-IHHO-BiLSTM

The data are selected from Table 1. Figure 8 shows the comparison of the iteration curves of HHO and DE-IHHO. Due to the initialization of the population by the chaotic dyadic learning strategy, the adaptation value of DE-IHHO is much smaller than that of HHO at the initial stage. Due to the introduction of the mutation operation strategy, the parallel optimization mechanism, and the smoothing process, DE-IHHO can quickly converge to a more accurate value later. In summary, DE-IHHO has higher convergence accuracy and convergence speed than HHO.

The results of the hyperparameter optimization selection are shown in Table 5. The number of training rounds is changed to the original 200 rounds, the learning rate is changed to 0.0008, and the number of neurons in the hidden

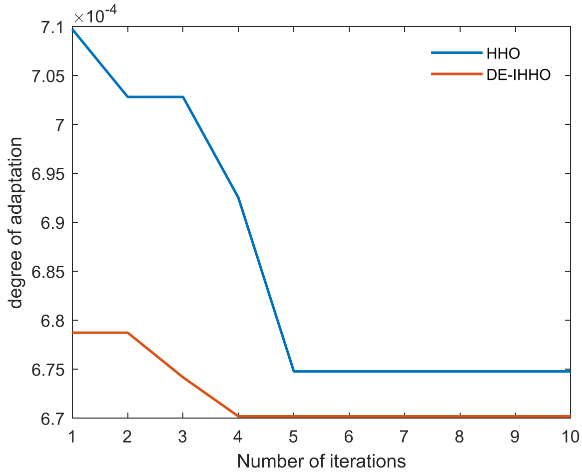


FIGURE 8. Comparison of HHO and DE-IHHO Performance.

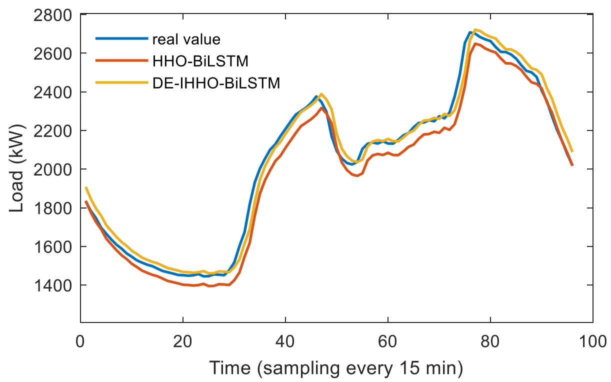


FIGURE 9. Comparison of prediction results of the DE-IHHO-BiLSTM model.

layer is changed to 12. The prediction results are shown in Fig. 9, and the error is reduced to 888.2 KW·h, accounting for 1.8% of the total daily electricity consumption, compared to the error of the unoptimized BiLSTM is reduced by 3.2%. As shown in Table 6, the improved DE-IHHO-BiLSTM improves the MAE, MAPE, and RMSE by 24.39KW, 1.23%, and 26.82KW, respectively compared to the HHO-BiLSTM model, which is a significant improvement in prediction, indicating that the hyper-parameters selected by DE-IHHO dramatically improves the prediction performance of the model.

V. CONCLUSION

To improve the accuracy of grid load forecasting to ensure the smooth progress of grid scheduling and user-side demand response, this paper finally constructs the DE-IHHO-BiLSTM forecasting model, and the following conclusions are obtained through comparative analysis:

(1) Construct BiLSTM load prediction model, compared with other prediction models BiLSTM with bi-directional structure can better capture the features in the time-series data, and through the experiment proved that the prediction results of BiLSTM are closer to the real value, due to the hyper-parameters are given empirically is not optimal, resulting in larger prediction error.

(2) Aiming at the problem of difficult selection of hyper-parameters for BiLSTM, this paper applies HHO, which has strong local search ability, to select the optimal hyperparameters for BiLSTM to improve load prediction accuracy. The experimental results show that the prediction accuracy of HHO-BiLSTM is significantly enhanced for BiLSTM. However, the global search ability of HHO is insufficient, and better hyperparameter combinations are still to be explored.

(3) To improve the global search capability of HHO to get better hyperparameters, this paper designs a hybrid parallel algorithm of HHO and DE to enhance the diversity of search and adopts a chaotic dyadic learning strategy and mutation operation strategy to improve the global search capability of HHO, and finally smoothes the optimal solution to reduce the influence of abnormal solution. The experimental results show that the improved DE-IHHO has better convergence speed and optimality searching ability than HHO, and better hyper-parameter combinations are obtained to improve further the prediction accuracy, which indicates that the combined DE-IHHO-BiLSTM prediction model has a specific application value.

REFERENCES

[1] F. Liu and C. Liang, "Short-term power load forecasting based on AC-BiLSTM model," *Energy Rep.*, vol. 11, pp. 1570–1579, Jun. 2024, doi: 10.1016/j.egyr.2024.01.026.

[2] B. Li and M. H. Lu, "Short-term load forecasting modeling of regional power grid considering real-time meteorological coupling effect," *Autom. Electr. Power Syst.*, vol. 44, no. 17, pp. 60–68, 2020.

[3] H. Chen, M. Zhu, X. Hu, J. Wang, Y. Sun, J. Yang, B. Li, and X. Meng, "Multifeature short-term power load forecasting based on GCN-LSTM," *Int. Trans. Electr. Energy Syst.*, vol. 2023, pp. 1–10, Aug. 2023, doi: 10.1155/2023/8846554.

[4] J. F. Rendon-Sanchez and L. M. de Menezes, "Structural combination of seasonal exponential smoothing forecasts applied to load forecasting," *Eur. J. Oper. Res.*, vol. 275, no. 3, pp. 916–924, Jun. 2019.

[5] P. Y. Chen and Y. J. Fang, "Short-term load forecasting of power system for holiday point-by-point growth rate based on Kalman filtering," *Eng. J. Wuhan Univ.*, vol. 53, no. 2, pp. 139–144, 2020.

[6] N. Kandil, R. Wamkeue, M. Saad, and S. Georges, "An efficient approach for short term load forecasting using artificial neural networks," *Int. J. Electr. Power Energy Syst.*, vol. 28, no. 8, pp. 525–530, Oct. 2006, doi: 10.1016/j.ijepes.2006.02.014.

[7] J. B. Pollack, "Recursive distributed representations," *Artif. Intell.*, vol. 46, nos. 1–2, pp. 77–105, Nov. 1990.

[8] A. I. Arvanitidis, D. Bargiotas, A. Daskalopulu, V. M. Laitsos, and L. H. Tsoukalas, "Enhanced short-term load forecasting using artificial neural networks," *Energies*, vol. 14, no. 22, p. 7788, Nov. 2021, doi: 10.3390/en14227788.

[9] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.

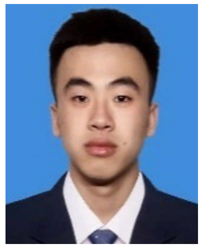
[10] Z. Sheng, Z. An, H. Wang, G. Chen, and K. Tian, "Residual LSTM based short-term load forecasting," *Appl. Soft Comput.*, vol. 144, Sep. 2023, Art. no. 110461, doi: 10.1016/j.asoc.2023.110461.

[11] Y. Zeng, H. H. Yang, Y. S. Feng, Z. Wang, D. Y. Zhao, and Y. Zeng, "A convolution BiLSTM neural network model for Chinese event extraction," in *Natural Language Understanding and Intelligent Applications*, vol. 10102. Cham, Switzerland: Springer, 2016, pp. 275–287.

[12] M.-P. Wu and F. Wu, "Predicting residential electricity consumption using CNN-BiLSTM-SA neural networks," *IEEE Access*, vol. 12, pp. 71555–71565, 2024, doi: 10.1109/access.2024.3400972.

[13] J. Yang, X. Zhang, W. Chen, and F. Rong, "Prophet-CEEMDAN-ARBiLSTM-based model for short-term load forecasting," *Future Internet*, vol. 16, no. 6, p. 192, May 2024, doi: 10.3390/fi16060192.

[14] L. Cai, H. C. Zhou, H. Bai, Z. G. Cai, and K. T. Yin, ''Application load forecasting method based on multilayer bi-directional LSTM and improved PSO algorithm,'' *Zhejiang Daxue Xuebao Gongxue Ban/J. Zhejiang Univ. Eng. Sci.*, vol. 54, no. 12, pp. 2414–2422, 2020.
[15] Q. Ge, C. Guo, H. Jiang, Z. Lu, G. Yao, J. Zhang, and Q. Hua, ''Industrial power load forecasting method based on reinforcement learning and PSO-LSSVM,'' *IEEE Trans. Cybern.*, vol. 52, no. 2, pp. 1112–1124, Feb. 2022, doi: 10.1109/TCYB.2020.2983871.
[16] Y. P. Shi, L. Qi, and T. L. Yong, ''Multivariable sales prediction for filling stations via GA improved BiLSTM,'' *Petroleum Sci.*, vol. 19, no. 5, pp. 2483–2496, 2022.
[17] A. S. Santra and J.-L. Lin, ''Integrating long short-term memory and genetic algorithm for short-term load forecasting,'' *Energies*, vol. 12, no. 11, p. 2040, May 2019, doi: 10.3390/en12112040.
[18] M. Sheikhan and N. Mohammadi, ''Neural-based electricity load forecasting using hybrid of GA and ACO for feature selection,'' *Neural Comput. Appl.*, vol. 21, no. 8, pp. 1961–1970, Nov. 2012, doi: 10.1007/s00521-011-0599-1.
[19] R.-A. Hooshmand, H. Amooshahi, and M. Parastegari, ''A hybrid intelligent algorithm based short-term load forecasting approach,'' *Int. J. Electr. Power Energy Syst.*, vol. 45, no. 1, pp. 313–324, Feb. 2013, doi: 10.1016/j.ijepes.2012.09.002.
[20] L. L. Peng, G. F. Fan, M. L. Huang, and W. C. Hong, ''Hybridizing DEMD and quantum PSO with SVR in electric load forecasting,'' *Energies*, vol. 9, no. 3, p. 221, 2016, doi: 10.3390/en9030221.
[21] T. Bashir, C. Haoyong, M. F. Tahir, and Z. Liqiang, ''Short term electricity load forecasting using hybrid prophet-LSTM model optimized by BPNN,'' *Energy Rep.*, vol. 8, pp. 1678–1686, Nov. 2022, doi: 10.1016/j.egyr.2021.12.067.
[22] Q. Sun and H. Cai, ''Short-term power load prediction based on VMD-SG-LSTM,'' *IEEE Access*, vol. 10, pp. 102396–102405, 2022, doi: 10.1109/ACCESS.2022.3206486.
[23] S. H. Rafi, Nahid-Al-Masood, S. R. Deeba, and E. Hossain, ''A short-term load forecasting method using integrated CNN and LSTM network,'' *IEEE Access*, vol. 9, pp. 32436–32448, 2021, doi: 10.1109/ACCESS.2021.3060654.
[24] X. Qi, H. Wang, Y. Ji, Y. Li, X. Luo, R. Nie, and X. Liang, ''Daily natural gas load prediction method based on APSO optimization and attention-BiLSTM,'' *PeerJ Comput. Sci.*, vol. 10, p. e1890, Feb. 2024, doi: 10.7717/peerj-cs.1890.
[25] H. Gezici and H. Livatyalı, ''Chaotic Harris hawks optimization algorithm,'' *J. Comput. Design Eng.*, vol. 9, no. 1, pp. 216–245, Feb. 2022, doi: 10.1093/jcde/qwab082.
[26] K. Hussain, W. Zhu, and M. N. M. Salleh, ''Long-term memory Harris' hawk optimization for high dimensional and optimal power flow problems,'' *IEEE Access*, vol. 7, pp. 147596–147616, 2019, doi: 10.1109/ACCESS.2019.2946664.
[27] X. Li, S. Ma, and J. Hu, ''Multi-search differential evolution algorithm,'' *Int. J. Speech Technol.*, vol. 47, no. 1, pp. 231–256, Jul. 2017, doi: 10.1007/s10489-016-0885-9.
[28] L. Fu, H. Zhu, C. Zhang, H. Ouyang, and S. Li, ''Hybrid harmony search differential evolution algorithm,'' *IEEE Access*, vol. 9, pp. 21532–21555, 2021, doi: 10.1109/ACCESS.2021.3055530.
[29] L. Zhu, Y. Ma, and Y. Bai, ''A self-adaptive multi-population differential evolution algorithm,'' *Natural Comput.*, vol. 19, no. 1, pp. 211–235, Mar. 2020, doi: 10.1007/s11047-019-09757-3.
[30] S. Yazdani and E. Hadavandi, ''LMBO-DE: A linearized monarch butterfly optimization algorithm improved with differential evolution,'' *Soft Comput.*, vol. 23, no. 17, pp. 8029–8043, Sep. 2019, doi: 10.1007/s00500-018-3439-8.
[31] H. Wang, J. Tang, and Q. Pan, ''MSI-HHO: Multi-strategy improved HHO algorithm for global optimization,'' *Mathematics*, vol. 12, no. 3, p. 415, Jan. 2024, doi: 10.3390/math12030415.
[32] M. Han, J. Zhong, P. Sang, H. Liao, and A. Tan, ''A combined model incorporating improved SSA and LSTM algorithms for short-term load forecasting,'' *Electronics*, vol. 11, no. 12, p. 1835, Jun. 2022, doi: 10.3390/electronics11121835.



XUELEI LIU was born in Liaoning, China, in 1999. He is currently pursuing the master's degree with Shenyang Agricultural University. His research interest includes power load forecasting.

ZIQI MA was born in Hebei, China, in 2001. She is currently pursuing the master's degree with Shenyang Agricultural University. Her research interests include image processing and target detection.

HANRUI GUO was born in Shanxi, China, in 2002. She is currently pursuing the master's degree with Shenyang Agricultural University. Her research interest includes control engineering.

YEDONG XU was born in Liaoning, China, in 2001. He is currently pursuing the master's degree with Shenyang Agricultural University. His research interests include image processing and target detection.

YINGLI CAO received the Ph.D. degree in information and electrical engineering from Shenyang Agricultural University, in 2010. She is currently a Professor and a Doctoral Supervisor with Shenyang Agricultural University. Her research interest includes agricultural informatisation.